COMS 4705: Intro to NLP

Assignment 2

Columbia University, John Hewitt

Instructions: Your responses to each question must be **written in latex** and uploaded to gradescope as a **pdf**. In gradescope be sure to select the correct pages for each question when you submit (those selected pages are what a TA is able to see while grading, if they are unselected the TA sees no response to the question). Ask questions on Ed as the come up!

Problem 1 (2 points) *Content from lecture 3*

After introducing attention weights computed from token similarity, Lecture 3 describes this setup as "*Not perfect, but better!*" Recall that in this mechanism, token i draws information from earlier tokens with weights

$$\alpha_{ij} \propto \exp\Big((Ex_i)^\top (Ex_j)\Big), \qquad h_{<i} = \sum_{j<i} \alpha_{ij}\, Ex_j.$$

For each part below, build a short prefix using only words from this bank (words may be reused), and consider the representation of the prefix's *last* token:

```
the   we   did   not   stopped   stop   train   at   station
```

(a) Build a prefix where attention is **better** than the plain average $\frac{1}{i-1}\sum_{j<i} Ex_j$: the words the last token needs are the ones it will attend to. Explain why, using how α_{ij} is computed.

(b) Build a prefix where attention is **not perfect**: the last token needs a word it will *not* attend to. Explain why, using how α_{ij} is computed.

Problem 2 (3 points) *Content from lecture 4*

Let $h_\ell \in \mathbb{R}^d$ denote the representation of a single token at layer ℓ, and let $f : \mathbb{R}^d \to \mathbb{R}^d$. Consider a residual update $h_\ell \leftarrow h_{\ell-1} + f(h_{\ell-1})$ and a gated update $h_\ell \leftarrow h_{\ell-1} + g \odot f(h_{\ell-1})$, where $g \in [0,1]^d$.

(a) What value of g recovers the plain residual update? Write one sentence on what this says about the relationship between residual connections and gates. (1 point)

(b) Suppose g is computed as $\sigma(Wh_{\ell-1} + b)$, where $W \in \mathbb{R}^{d\times d}$ and $b \in \mathbb{R}^d$ are learned parameters and σ is the elementwise sigmoid. The bias b is initialized very negative, so that $g \approx 0$ for every input. What does a stack of many such blocks compute at initialization, and why is this a desirable starting point for a deep network? (2 points)

Problem 3 (3 points) *Content from lecture 5*

Consider the sentence, tokenized by whitespace with tokens indexed from 0:

0	1	2	3	4	5	6	7	8	9	10	11
`Zuko`	`and`	`Uncle`	`Iroh`	`opened`	`a`	`tea`	`shop`	`in`	`Ba`	`Sing`	`Se`

A span (s, e) covers tokens s through $e - 1$ (start inclusive, end exclusive), so $(2, 4)$ is *Uncle Iroh*. The gold (human-annotated) named-entity spans are

$$y = \{\, (0, 1)\ \textit{Zuko},\ (2, 4)\ \textit{Uncle Iroh},\ (9, 12)\ \textit{Ba Sing Se}\,\}.$$

A system predicts

$$\hat{y} = \{\, (0, 1)\ \textit{Zuko},\ (3, 4)\ \textit{Iroh},\ (6, 8)\ \textit{tea shop},\ (9, 12)\ \textit{Ba Sing Se}\,\}.$$

(a) Compute precision, recall, and F1 for $\hat{y}$ using the definitions from lecture.

(b) Describe a system that achieves recall of 1 on this sentence, and give its precision and F1.

(c) Describe a system that achieves precision of 1 on this sentence, and give its recall and F1. Together with (b), what does this tell you about reporting precision or recall alone?

Problem 4 (12 points) *Content from lectures 3 and 4*

Complete the code provided in `a2.ipynb`. Answer the short questions in the notebook and write the closing discussion. Upload the completed notebook to Gradescope.